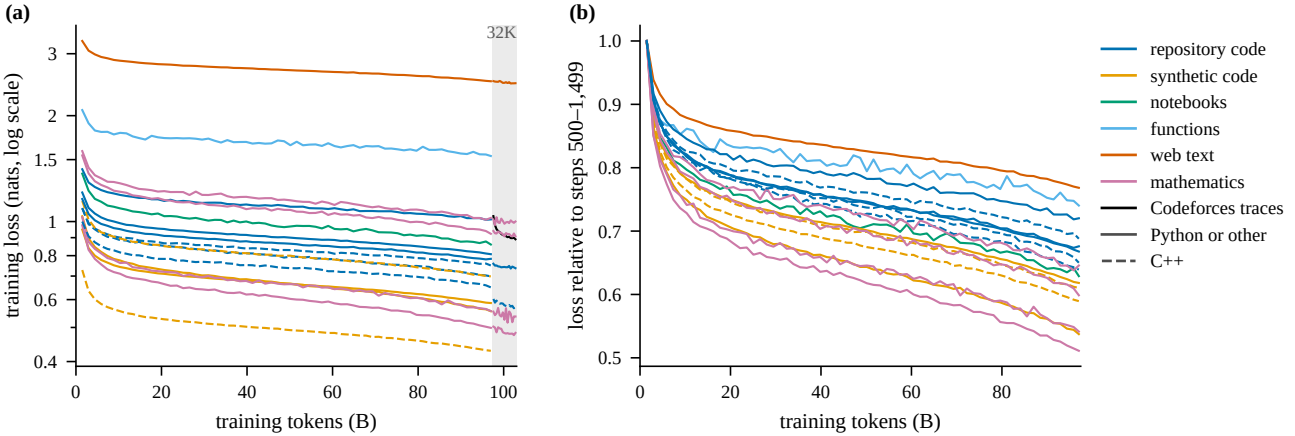
(a)
32K
training loss (nats, log scale)
3
2
1.5
1
0.8
0.6
0.4
0
20
40
60
80
100
training tokens (B)
(b)
loss relative to steps 500–1,499
1.0
0.9
0.8
0.7
0.6
0.5
0
20
40
60
80
training tokens (B)
repository code
synthetic code
notebooks
functions
web text
mathematics
Codeforces traces
Python or other
C++